Forecasting US State Unemployment Rates Three Months Ahead

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GitHub: https://github.com/spcddou/outlier-data1030-fall2026

Introduction

Using a state's current and lagged unemployment rates, we plan to predict its unemployment rate three reference months ahead. If the forecasts are sufficiently accurate, state employment agencies can plan staffing and service resources in advance. At this stage, we collect and check the data and use exploratory data analysis to examine unemployment-rate patterns.

Previous research has shown that the information available at the time of forecasting and the version of data used can both influence the results. The FRED-SD study by Bokun et al. (2024) forecast state-level unemployment rates, including a three-month horizon, and found that real-time data versus revised historical data can alter model comparison results. Şahin et al. (2021) used labor-market flow data to study national unemployment rate forecasts during the pandemic. Aaronson et al. (2020) used job-search activity to forecast state-level unemployment insurance claims.

Differences in data, targets, and methods prevent direct benchmark comparisons.

Data (A complete description of all variables is provided in Appendix A)

We sent six batches of JSON POST requests through the Public Data API maintained by the U.S. Bureau of Labor Statistics (BLS, 2014) to retrieve seasonally adjusted monthly unemployment rates for the 50 states and the District of Columbia from the Local Area Unemployment Statistics program. The data cover the period from January 2014 to December 2024. The raw responses, request parameters, and UTC collection times were all preserved. The

data was collected on the evening of October 14, 2026, Eastern Time, corresponding to October 15 in UTC.

The raw data table contains 6,732 rows and 3 columns, including state codes, months, and unemployment rates. Each row represents one month for a given region. Unemployment rates are expressed as percentages and are defined as the ratio of the number of unemployed persons to the labor force (BLS, 2017).

We checked for duplicate state-month records, incomplete monthly coverage, invalid rates, and annual-average observations. All 51 regions contain 132 monthly observations, with no missing or duplicate state-month records.

We aligned observations to a complete monthly calendar within each state and paired the current unemployment rate with the actual rate for the same state three reference months later. For exploratory reference months from January 2015 to September 2024, the complete paired table has 5,967 rows and 5 columns: state code, reference month, current unemployment rate, future unemployment rate, and target month. These pairs are used for EDA; no model is fitted.

Observations from 2014 are retained in the source panel. The EDA focuses on 2015–2024, and the last three months of 2024 lack corresponding future labels and are therefore excluded from the paired table. The observations are not independent and identically distributed (i.i.d.). Observations within a state may be serially correlated, and states may experience common shocks.

Exploratory Data Analysis

The first figure shows the changes in unemployment rates across states over time. Many regions exhibited a downward trend prior to 2020, followed by a sharp increase around 2020 and

a gradual decline thereafter. The magnitude of change varies across states, and the model must account for both relatively stable and volatile periods.

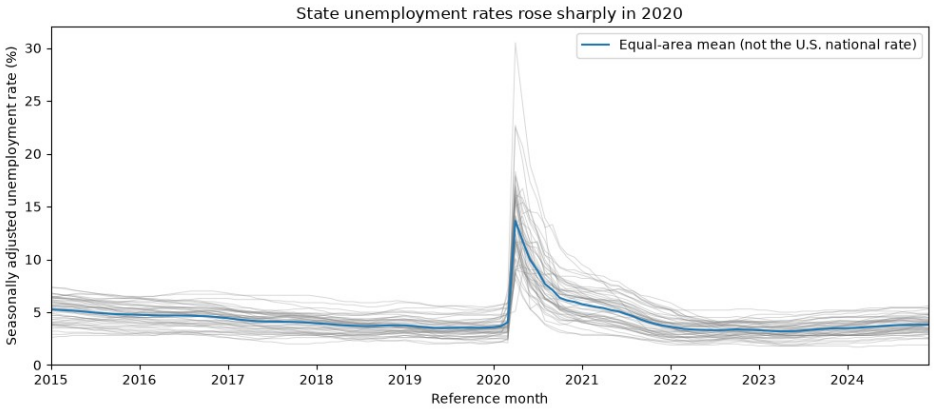


Figure 1: Seasonally adjusted monthly unemployment rates across 50 states and the District of Columbia, 2015–2024. Gray lines represent individual areas, while the blue line represents the equal-area mean rather than the official U.S. unemployment rate. Source: BLS API data used in this study.

The second figure compares the current unemployment rate with the actual rate three reference months later. Among all 5,967 complete pairs, 86.0% of observations showed a change of no more than 0.5 percentage points within three months. State-level unemployment rates typically exhibit strong short-term persistence, and the data also contain observations that lie far from the equality line. The colors identify forecast-origin periods only and do not represent causal effects of the pandemic.

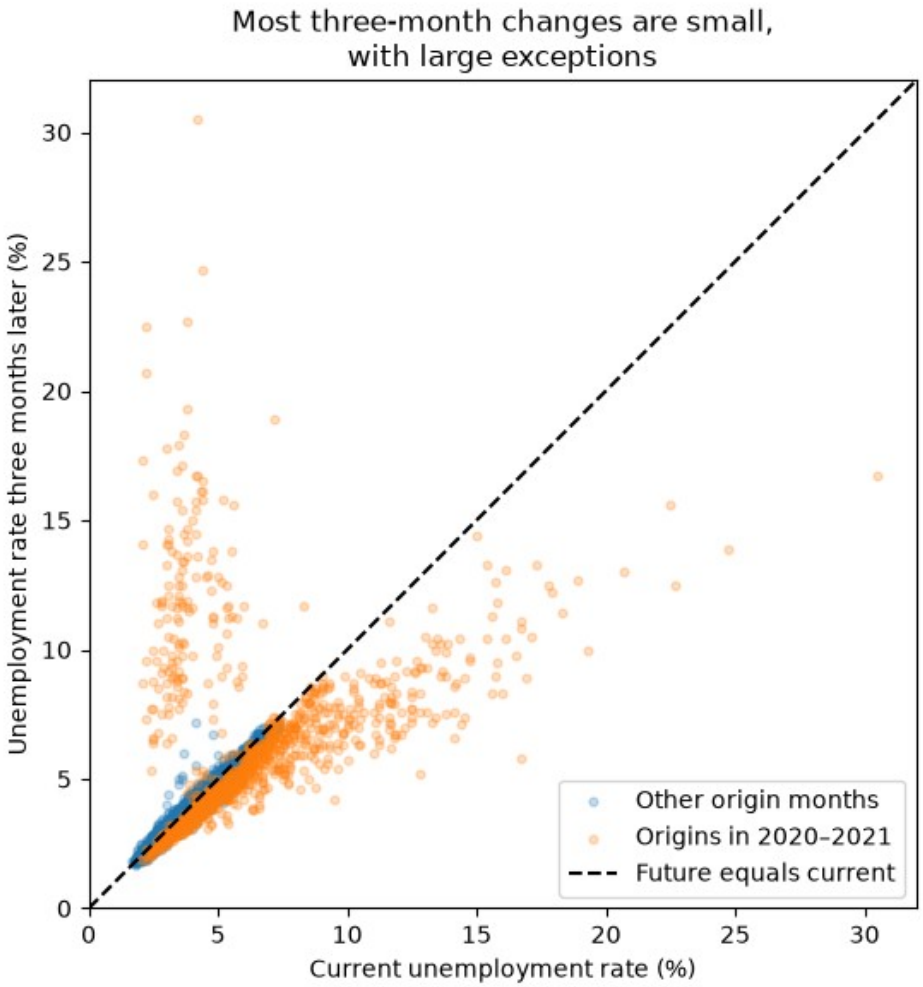


Figure 2: Current Unemployment Rate vs. Actual Unemployment Rate Three Reference Months Later. Each data point represents a state-month pair; the dashed line indicates equality between the two. This figure does not compare model predictions with actual values. Source: BLS API data from this dataset.

The third figure compares the distributions of the future unemployment rate across calendar periods. For reference months in 2020–2021, the future unemployment rate ranged from 1.9% to 30.5%, while the 2022–September 2024 group ranged from 1.7% to 5.6%. These calendar periods cover different unemployment distributions; they are descriptive groups, not training and test sets.

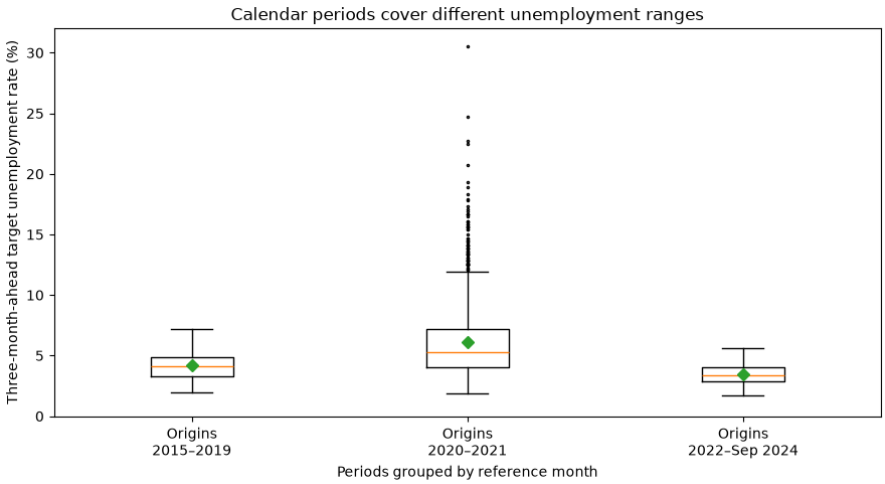


Figure 3: Distribution of future unemployment rates across different periods. Boxes represent the median and interquartile range; whiskers extend to the outermost observations within 1.5 times the interquartile range; dots represent observations outside this range; and diamonds represent the mean. Source: This set of BLS API data.

Figure 4 examines autocorrelation separately for all 50 states and DC using 120 months per area, January 2015–December 2024. We use the mean-centered definition underlying pandas' autocorrelation plot from Lecture 10. Short-lag autocorrelation is generally positive and declines with lag, although magnitudes differ across areas. This supports examining historical rates as potential predictors; it does not establish forecast accuracy or select optimal lags.

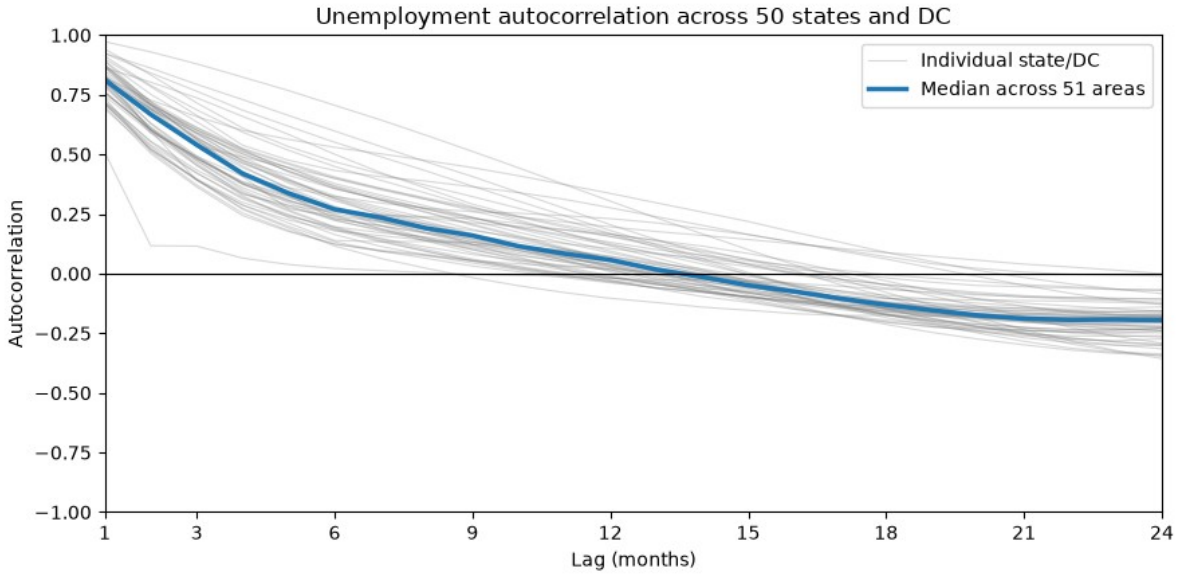


Figure 4: Unemployment-rate autocorrelation across 50 states and DC, January 2015–December 2024. Gray lines show each area's autocorrelation; the blue line is the median across areas at each monthly lag. The median is not the autocorrelation of an average unemployment series. Source: our BLS API data.

The full 2015–2024 period has already been explored and is not an untouched test set. Any subsequent model development will need time-ordered validation and an explicit account of which periods have already been inspected.

Problem description and potential impact

This project is a supervised regression task using panel time-series data, with monthly observations for 50 states and the District of Columbia. We aim to forecast each state's unemployment rate three reference months ahead using its historical unemployment rates and state identity. Observations are ordered in time, and rates within each state may be serially correlated, as explored in Figure 4. Month t represents the reference month to which the latest published data belongs, and the target is t plus 3 months. State-level data are generally released in the month following the reference month (BLS, 2025); therefore, “three months after the

reference month" does not exactly correspond to "three months from the release date." The current code does not reconstruct the exact historical release dates for each month.

This API snapshot contains revised historical data and does not represent the initially released data for each month. The BLS conducts monthly and annual revisions; annual historical smoothing may also utilize information from periods after the target month (BLS, 2018, 2025). Constructing lagged variables based on time alone does not prove that all source data were available at the time. The current work is exploratory analysis using revised historical data, and the impact of data revisions on future forecast errors has not yet been quantified.

Next, we plan to use ordinary linear regression with lagged unemployment rates and state indicators for three-month-ahead forecasting. Validation will preserve chronological order, with earlier observations used for training and later observations for validation. No model has been trained or evaluated in this midterm notebook.

References

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Appendix A

The accompanying local data/data_dictionary.md documents the source panel and the EDA paired table.

Appendix B AI Assistance Disclosure

OpenAI ChatGPT was used to review English grammar, improve wording and readability, and provide formatting suggestions.

AI interaction record: https://chatgpt.com/s/t_6ada2f46e8b08191b2131f57ea51a6a2

https://chatgpt.com/s/t_6adcefe420c4819194e852fc1fead05f

https://chatgpt.com/s/t_6adcf0658eb48191986532c03cd6966c